

Introduction to Contract Theory and Auctions  
Lecture 4  
Lotteries with Monetary Prizes & Applications

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- Our discussion begin where we left last time: a DM satisfies the VNM assumptions where the space of prizes is now a set  $Z \subseteq \mathbb{R}$  and  $a \in Z$  is interpreted as receiving \$ $a$ .
- In contrast to the last lecture in which we assumed the set  $Z$  was finite, in this lecture we will assume that it is an infinite set.
- Due to technical reasons we will focus on a restricted set of possible lotteries:
  - For some examples, we will use the set of lotteries which are **finitely supported** on  $Z$ , i.e. if  $p$  is a lottery, then  $\text{supp}(p) \subseteq \mathbb{R}$  is a finite set of points.
  - However, for most of the lecture we will generalize our analysis to the set of lotteries consists of lotteries which are supported on a compact<sup>1</sup> subset of  $\mathbb{R}$ .
    - For many of the proofs, we will make the extra assumption that if  $p$  is a lottery, it has a well defined probability density function, i.e.  $p$  is a pdf.

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<sup>1</sup>A compact subset of  $\mathbb{R}$  is closed and bounded.

- Let  $Z$  be an **infinite** set of prizes,  $Z = [w, b] \subseteq \mathbb{R}$ ,  $w < b$ .
- Let  $\mathfrak{F}$  be the set of CDFs on  $Z$ .
- Recall the following definition.

### Definition 4.1

For any  $F \in \mathfrak{F}$ , the following conditions hold:

- ①  $\lim_{x \rightarrow w} F(x) = 0$  and  $\lim_{x \rightarrow b} F(x) = 1$ .
  - ②  $F(x)$  is a nondecreasing function of  $x$ .
  - ③  $F(x)$  is right-continuous; that is, for every number  $x_0$ ,  $\lim_{x \downarrow x_0} F(x) = F(x_0)$ .
- The set  $\mathfrak{F}$  will play the role of the set of lotteries.

- Suppose  $\psi : Z \rightarrow \mathbb{R}$  is a bounded function. For a  $F \in \mathfrak{F}$ , consider the following integral:

$$\int_Z \psi(z) dF(z), \quad (1)$$

the previous integral is in the sense of *Lebesgue–Stieltjes* and should be interpreted as the **expectation of  $\psi$  with respect to  $F$** .

- The *Lebesgue–Stieltjes* integral generalizes the integral you are probably most familiar with, the *Riemann Integral*.
- Using the integral in (1) allows us to integrate over a larger class of functions, but most importantly, consolidates under the same concepts the “discrete” and “continuous” cases.
  - If  $p$  is a lottery with finite support in  $Z$ , i.e., it attaches positive probability  $p(z) > 0$  to the finite collection of prizes  $z_k \in Z$ ,  $k = 1, \dots, K$ , then  $F(x) := \sum_{z_k \in Z: z_k \leq x} p(z_k)$  and the integral in (1) simplifies to

$$\int_Z \psi(z) dF(z) = \sum_{z \in Z} \psi(z) p(z)$$

- Given  $F \in \mathfrak{F}$ , we can define its corresponding “lottery” as follows: for a subset of prizes  $A \subseteq Z$ , the probability that lottery  $p$  attaches to the prize  $z$  being in  $A$  is defined as:

$$p(A) := \Pr(z \in A) = \int_Z \mathbf{1}_A(z) dF(z), \quad (2)$$

where  $\mathbf{1}_A$  denotes the indicator function of the set  $A$ .

- Thus, the lottery  $p$  is a function defined over subsets of  $Z$  that attaches a probability to each subset.
- Not all subsets of  $Z$  can be assigned a probability; only a certain subclass of subsets can be assigned a “meaningful” number that can be interpreted as a probability. I am not being rigorous here on purpose, and I will brush aside this technical point to not complicate more the discussion.
- Note that in definition 4.1 it is not required for the cdf  $F$  to be fully continuous, just right continuous.

- However, in the case that the cdf  $F$  is not only right-continuous but satisfies a strong notion of continuity (absolute continuity), then we have the following definition:

### Definition 4.2

The **probability density function** or pdf,  $f(x)$  of the cdf  $F \in \mathfrak{F}$  is the function that satisfies

$$F(x) = \int_w^x f(t)dt \quad \text{for all } x$$

- Thus, if  $F \in \mathfrak{F}$  has a pdf  $f$ , then the integral in (1) can be simplified as

$$\int_w^b \psi(z)f(z)dz, \tag{3}$$

where the integral in (3) is the standard Riemann integral that you know from Calculus I.

We adjust the vNM axioms slightly to fit this framework. Consider a binary relation  $\succsim$  over  $\mathfrak{F}$ .

### Definition 4.3 (Completeness and Transitivity (A) + Monotonicity for Money (MM))

$\succsim$  is complete and transitive, i.e. a preference relation and it satisfies monotonicity for money, that is  $x > y$  implies  $[x] \succsim [y]$ .

### Definition 4.4 (Independence (I))

$p \succ q$ , and  $\alpha \in (0, 1)$  and implies that  $\alpha p \oplus (1 - \alpha)r \succ \alpha q \oplus (1 - \alpha)r$  for any  $r \in \mathfrak{F}$ .

We will of course need some form of continuity axiom. The form of continuity we impose is stronger than the one we saw before.

### Definition 4.5 (Continuity (C))

For every  $F \in \mathfrak{F}$ , the sets  $\{G : G \succsim F\}$  and  $\{G : F \succsim G\}$  are closed under the topology induced by the metric

$$d(F, G) = \int_{\mathbb{R}} |F(x) - G(x)| dx$$

- (C) implies that we are endowing  $\mathfrak{F}$  with the weak-topology.
- This is the topology of convergence in distribution, e.g. if  $F_n \rightarrow F$  at every continuity point of  $F$  and  $F_n \succsim G$  for all  $n$ , then  $F \succsim G$ .

### Example 4.6

- Consider the sequence of sure lotteries  $[1 + 1/n]$ ,  $n \in \mathbb{N}$ . As you can easily see, the cdfs converge to the cdf of the lottery  $[1]$ .
- Consider the sequence of lotteries  $p_n$  which gives prizes  $\{0, 1/n, 2/n, \dots, (n-1)/n, 1\}$ , each with probability  $1/(n+1)$ . This sequence converges in distribution to the uniform distribution.
- Consider now the sequence of lotteries whose cdf's are the normal cdf's with mean 4 and variance  $1/n$ . This sequence converges in distribution to the degenerate distribution at value 4.

### Theorem 4.7

The binary relation  $\succsim$  satisfies A, MM, I and C if and only if there is a continuous, strictly increasing  $u : Z \rightarrow \mathbb{R}$ , such that

$$U(F) = \int_Z u(z) dF(z) \quad \text{represents } \succsim.$$

- The function  $u$  is called the vNM utility index, or Bernoulli utility function. The function  $U$  is the expected utility.
- I will prove this theorem, since the proof is advanced.<sup>2</sup>

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<sup>2</sup>However, if you are interested in taking this to a greater depth and understanding the gory details, the following references are useful:

- David M. Kreps. *Microeconomic Foundations I: Choice and Competitive Markets*. Princeton University Press, 2013. This is a good place to start.
- David M. Kreps. *Notes on the Theory of Choice*. Westview Press, 1988. If you want to go more in depth.
- Peter C. Fishburn. *Utility Theory for Decision Making*. John Wiley and Sons, 1970. If you want the full details. Difficult to find.

### Definition 4.8 (Attitudes towards Risk)

Let  $\succsim$  be a preference relation over  $\mathfrak{F}$ .

- We say that  $\succsim$  is **risk averse** if for any lottery  $p$ ,  $[E(p)] \succsim p$ , i.e. if the DM prefers the sure lottery that gives as a prize the expected value of the lottery versus receiving the lottery itself.
- We say that  $\succsim$  is **risk neutral** if for any lottery  $p$ ,  $[E(p)] \sim p$ .
- We say that  $\succsim$  is **risk loving** if for any lottery  $p$ ,  $[E(p)] \precsim p$ .

Note that if the cdf  $F$  represents a lottery and preferences admit an expected utility representation with Bernoulli utility function  $u(z)$  a DM is risk averse if and only if

$$\int_Z u(z) dF(z) \leq u\left(\int_Z z dF(z)\right). \quad (4)$$

## Risk aversion is equivalent to concavity of $u$

- Note that the inequality in 4 is a defining property of concave functions, i.e. if preferences are representable a Bernoulli utility function  $u$ , then **risk aversion is equivalent to the concavity of  $u$** .
- Recall some basic properties of concave functions:
  - An increasing and concave function must be continuous (but not necessarily differentiable.)
  - Jensen's inequality*. If  $u$  is concave, then for any finite collection of numbers  $(\alpha_k)_{k=1,\dots,K}$  that add up to 1,  $u\left(\sum_{k=1}^K \alpha_k x_k\right) \geq \sum_{k=1}^K \alpha_k u(x_k)$ . Note that equation (4) is the “continuous” version of Jensen's inequality, where the weights  $\alpha$  have been replaced by the CDF,  $F$ . You can see this more easily if you assume that  $F$  has a pdf  $f$ .
  - Three Strings Lemma*: For any  $a < b < c$ , we have

$$\frac{u(c) - u(b)}{c - b} \leq \frac{u(c) - u(a)}{c - a} \leq \frac{u(b) - u(a)}{b - a}.$$

- If  $u$  is twice differentiable, then for any  $a < c$ ,  $u'(a) \geq u'(c)$ , and thus  $u''(x) \leq 0$  for all  $x$ .

## Definition 4.9 (Certainty Equivalent and Risk Premium)

Given a Bernoulli utility function,  $u$ , we define the following concepts:

- 1 The **certainty equivalent** of a lottery  $F \in \mathfrak{F}$ , is the amount of money for which an individual is indifferent between  $F$  and the certain amount  $c(F, u)$ ; that is,  $c(F, u)$  is the number that solves the equation:

$$u(c(F, u)) = \int_Z u(z) dF(z)$$

- 2 The **risk premium** of a lottery  $F \in \mathfrak{F}$  is the difference

$$R(F, u) = E[F] - c(F, u),$$

where  $E[F]$  denotes the expected value of  $F$ , i.e.,  $E[F] = \int_Z z dF(z)$ .

## Proposition 4.10

*Suppose the decision maker's preferences are represented by an increasing  $u$ . The DM is a risk averter if and only if  $R(F, u) \geq 0$  for all  $F \in \mathfrak{F}$ .*

## Proof.

Let  $F \in \mathfrak{F}$ . Then we have the following chain of equivalent statements:

$$\begin{aligned} R(F, u) \geq 0 &\Leftrightarrow c(F, u) \leq \int z dF(z) \Leftrightarrow u(c(F, u)) \leq u\left(\int z dF(z)\right) \Leftrightarrow \\ &\int u(z) dF(z) \leq u\left(\int z dF(z)\right) \end{aligned}$$



## Application: Insurance

- Consider a strictly risk-averse decision maker who has an initial wealth of  $w$  but who runs a risk of a loss of  $D$  kzt.
- The probability of the loss is  $\pi$ .
- The DM can buy insurance. One unit costs  $q$  kzt and pays 1 kzt if the loss occurs.
- If  $\alpha$  units of insurance are bought, the wealth of the individual will be
  - $w - \alpha q$  if there is no loss
  - $w - \alpha q - D + \alpha$  if there is a loss.
- The expected wealth of the DM is then  $w - \pi D + \alpha(\pi - q)$ .
- The decision maker's problem is to choose the optimal level of  $\alpha$  :

$$\max_{\alpha \geq 0} (1 - \pi)u(w - \alpha q) + \pi u(w - \alpha q - D + \alpha)$$

- If  $\alpha^*$  is an optimum, it must satisfy the first-order condition:

$$-q(1 - \pi)u'(w - \alpha^* q) + \pi(1 - q)u'(w - D + \alpha^*(1 - q)) \leq 0$$

with equality if  $\alpha^* > 0$ .

## Actuarially fair insurance

- We are going to consider the case of **actuarially fair insurance**, that is, when the cost of insurance is equal to the probability of loss,  $q = \pi$ .
- Then the first order condition requires that

$$u'(w - D + \alpha^*(1 - \pi)) - u'(w - \alpha^*\pi) \leq 0$$

with equality if  $\alpha^* > 0$ .

- Since  $u'(w - D) > u'(w)$ , we must have  $\alpha^* > 0$ , and therefore

$$u'(w - D + \alpha^*(1 - \pi)) = u'(w - \alpha^*\pi)$$

- Because  $u'$  is strictly decreasing due to risk aversion, this implies

$$w - D + \alpha^*(1 - \pi) = w - \alpha^*\pi$$

which implies that  $\alpha^* = D$ .

- The lesson is: **if insurance is actuarially fair, the DM insures completely.**
- Alternative way: notice that if  $q = \pi$ , the DM's expected wealth is  $w - \pi D$  for any  $\alpha$ . Since setting  $\alpha = D$  allows him to reach  $w - \pi D$  with certainty, the definition of risk aversion implies that this is the optimal level of  $\alpha$ .

## Application: Demand for a Risky Asset

- An **asset** is a divisible claim to a financial return in the future.
- Suppose there are two assets, a safe asset with a return of 1 kzt per kzt invested and a risky asset with a random return of  $z$  kzt per kzt invested.
- The random return has a distribution  $F(z)$  that we assume satisfies  $\int z dF(z) > 1$ , so that its mean return exceeds the one of the safe asset.
- An individual has an initial wealth  $w$  to invest.
- Let  $\alpha$  and  $\beta$  denote the amounts of wealth invested in the risky and the safe asset, respectively. Thus, for any realization  $z$  of the random return, the individual's portfolio  $(\alpha, \beta)$  pays  $\alpha z + \beta$ .
- The budget restriction is  $\alpha + \beta = w$ .
- The individual must solve

$$\max_{\alpha, \beta \geq 0} \int u(\alpha z + \beta) dF(z)$$

such that  $\alpha + \beta = w$

- Alternatively, the individual's problem is

$$\max_{0 \leq \alpha \leq w} \int u(w + \alpha(z - 1)) dF(z)$$

- If  $\alpha^*$  is optimal, it must satisfy the first-order condition:

$$\phi(\alpha^*) = \int u'(w + \alpha^*(z - 1))(z - 1) dF(z) = \begin{cases} \leq 0 & \text{if } \alpha^* < w, \\ \geq 0 & \text{if } \alpha^* > 0. \end{cases}$$

- Note that  $\int z dF(z) > 1$  implies that  $\phi(0) > 0$ . Hence,  $\alpha^* = 0$  cannot satisfy this first-order condition.
- Lesson: if a risk is actuarially favorable, then a risk averter will always accept at least a small amount of it.

## Comparisons across individuals: the “More risk averse” relation & the coefficient of absolute risk aversion.

- We wish to formalize the statement “one decision maker is more risk averse than another”. Let’s first introduce the following definition.

### Definition 4.11

Given a (twice-differentiable) Bernoulli utility function  $u$  for money, the Arrow-Pratt coefficient of absolute risk aversion at  $x$  is defined as

$$r_A(x, u) = -\frac{u''(x)}{u'(x)}$$

- It turns out that  $r_A$  is a measure of risk aversion that is related to the curvature of  $u$ .
- We will go back to this point in a bit, but first I will state a proposition.

### Proposition 4.12

Consider two individuals utility functions  $u_1$  and  $u_2$ . Then, the following definitions of the *more-risk-averse-than* relation are equivalent: we say that  $u_1$  is more risk averse than  $u_2$  if:

- ①  $r_A(x, u_1) \geq r_A(x, u_2)$  for every  $x \in Z$ .
- ② There exist an increasing concave function  $\Psi$  such that  $u_1(x) = \Psi(u_2(x))$  at all  $x$ ; i.e.,  $u_1$  is a concave transformation of  $u_2$ .
- ③  $c(F, u_1) \leq c(F, u_2)$  for any  $F \in \mathfrak{F}$ .
- ④ Whenever  $u_1$  finds a lottery  $F$  at least as good as a risk-less lottery  $[c]$ , then  $u_2$  also finds  $F$  at least as good as  $[c]$ . That is  $\int u_1(z)dF(z) \geq u_1(c)$  implies  $\int u_2(z)dF(z) \geq u_2(c)$  for any  $F \in \mathfrak{F}$  and  $c \in Z$ .

In the homework, most likely I will ask you to prove the equivalence between (2), (3) and (4). In here I will prove that (1) and (2) are equivalent.

## Proof that (1) and (2) are equivalent.

Proof.

Start with  $u_1(x) = \Psi(u_2(x))$ . Differentiating we get

$$u_1'(x) = \Psi'(u_2(x))u_2'(x)$$

and

$$u_1''(x) = u_2''(x)\Psi'(u_2(x)) + \Psi''(u_2(x))(u_2'(x))^2$$

Dividing by  $-u_1'(x)$  and using the first expression, yields

$$r_A(x, u_1) = r_A(x, u_2) - \frac{\Psi''(u_2(x))}{\Psi'(u_2(x))} u_2'(x).$$

Thus  $r_A(x, u_1) \geq r_A(x, u_2)$  if and only if  $\Psi''(u_2) \leq 0$  for all  $u_2$  in the range of  $u_2 : Z \rightarrow \mathbb{R}$ . □

## Application: demand for a risky asset, continued.

- We analyze here the same asset portfolio problem as before.
- Suppose that we now have two individuals with Bernoulli utility functions  $u_1$  and  $u_2$ . Denote their optimal choices by  $\alpha_1^*$  and  $\alpha_2^*$ .
- We will show here that if  $u_1$  is more risk-averse than  $u_2$  then  $\alpha_1^* < \alpha_2^*$ ; i.e., the first decision maker invests less in the risky asset than the second one.
- Repeating from before, the problem of each decision maker  $i = 1, 2$  is

$$\max_{0 \leq \alpha \leq w} \int u_i(w + \alpha(z - 1)) dF(Z).$$

- Assuming an interior solution, we have that the optimal solutions for each  $i = 1, 2$  satisfy:

$$\phi_i(\alpha_i^*) = \int (z - 1) u_i'(w + \alpha_i^*(z - 1)) dF(z) = 0.$$

## Application: demand for a risky asset, continued.

- As we know, the concavity of  $u_1$  implies that  $\phi_1$  is decreasing. Therefore, if we show that  $\phi_1(\alpha_2^*) < 0$ , it must follow that  $\alpha_2^* > \alpha_1^*$ .
- Since  $u_1(x) = \Psi(u_2(x))$ , we can write

$$\phi_1(\alpha_2^*) = \int (z-1)\Psi'(u_2(w + \alpha_2^*(z-1)))u_2'(w + \alpha_2^*(z-1))dF(z) < 0.$$

- To understand the final inequality, note that the integrand in the expression is the same as in

$$\phi_2(\alpha_2^*) = \int (z-1)u_2'(w + \alpha_2^*(z-1))dF(z) = 0,$$

except that it is multiplied by  $\Psi'$ , which is a positive and decreasing function of  $z$ . (Recall that, by definition,  $\Psi$  is increasing and concave, thus  $\Psi'$  is positive and decreasing).

- Hence, the integral underweights the positive values of  $u_2'(w + \alpha_2^*(z-1))$ , which obtain for  $z > 1$ , relative to the negative values, which obtain for  $z < 1$ . Since in the last expression, the integral of the positive and the negative parts of the integrand added to zero, they must now add to a negative number in the former expression.

## Comparison of Payoff distributions in terms of return and risk.

- In the previous part of the lecture we focused on comparing utility functions.
- The aim now is to compare payoff distribution.
- In particular, we will try to answer questions of the type: “Given a set of preference relations on  $\mathfrak{F}$ , for what  $F, G \in \mathfrak{F}$  is  $F \succsim G$  for all preference relations in the set?”
- We will try to provide an answer to this question by formalizing some orders on the set  $\mathfrak{F}$ .

## First Order Stochastic Dominance

- We want to formalize the expression: “The distribution  $F$  yields unambiguously higher returns than the distribution  $G$ ”.
- There are two possibilities:
  - We could test whether every expected utility maximizer who values more over less money, prefers  $F$  to  $G$ .
  - Alternatively, we could verify if for every amount of money  $x$ , the probability of getting at least  $x$  is higher under  $F$  than  $G$ .
- Turns out that both ways of looking at this issue are the same.

### Definition 4.13

The distribution  $F \in \mathfrak{F}$  **first-order stochastically dominates**  $G$ , denoted  $F \triangleright_{\text{fosd}} G$ , if, for every non-decreasing function  $u : Z \rightarrow \mathbb{R}$ , we have

$$\int u(z) dF(z) \geq \int u(z) dG(z)$$

### Proposition 4.14

*Let  $F, G \in \mathfrak{F}$ . Then  $F \triangleright_{\text{fosd}} G$  if and only if  $F(x) \leq G(x)$  for every  $x$ .*

*Equivalently, define for any cdf  $F$ ,  $\bar{F}(z) = 1 - F(z)$  for every  $z \in Z$ . Intuitively,  $\bar{F}(z) = \Pr(\text{the monetary prize is greater than } z)$ . Then.  $F \triangleright_{\text{fosd}} G \Leftrightarrow$  if  $\bar{F}(x) \geq \bar{G}(x)$  for all  $x$ .*

## Proof about equivalence of FOSD definitions

### Proof.

Let  $F, G \in \mathfrak{F}$ . To simplify the proof, we will assume that the distributions  $F$  and  $G$  have pdf's  $f$  and  $g$  respectively. Note that it suffices to establish the result for differentiable utility functions  $u$ . Recall that we assumed  $Z = [w, b] \subset \mathbb{R}$ , where  $w < b$ . We establish an equivalent form for  $E[u(F)]$  first:

$$\begin{aligned}\int_w^b u(z) dF(z) &= \int_w^b u(z) f(z) dz = \int_w^b \left( \int_w^z u'(x) dx + u(w) \right) f(z) dz \\ &= \int_w^b \left( \int_w^z u'(x) dx \right) f(z) dz + u(w) = \int_w^b \int_x^b u'(x) f(z) dz dx + u(w) \\ &= \int_w^b u'(x) \left( \int_x^b f(z) dz \right) dx + u(w) = \int_w^b u'(x) \bar{F}(x) dx + u(w)\end{aligned}$$



### Proof.

We prove the  $\Leftarrow$  (if) part. Note that:

$$\int_w^b u(z) dF(z) \geq \int_w^b u(z) dG(z) \Leftrightarrow \int_w^b u'(x) \bar{F}(x) dx \geq \int_w^b u'(x) \bar{G}(x) dx.$$

Thus, if  $\bar{F}(x) \geq \bar{G}(x)$  and since  $u$  is increasing (and thus  $u' > 0$ ), then  $\int_w^b u'(x) \bar{F}(x) dx \geq \int_w^b u'(x) \bar{G}(x) dx$  and the if part of the proposition is true.

We now prove the  $\Rightarrow$  (only if) part. We use the contrapositive here. Suppose that there exists a prize  $\hat{z}$  for which  $\bar{F}(\hat{z}) < \bar{G}(\hat{z})$ . Define the following utility function  $\hat{u}(x) = 1$  for  $x > \hat{z}$  and  $\hat{u}(x) = 0$  for  $x \leq \hat{z}$ . This  $u$  is non-decreasing. But then  $\int_w^b \hat{u}(x) dF(x) = \bar{F}(\hat{z}) < \bar{G}(\hat{z}) = \int_w^b \hat{u}(x) dG(x)$ . Thus, by the contrapositive, the assertion is proved. □

## Example 4.15

Consider a compound lottery that has as its first stage a realization of  $x$  according to  $G$  and in its second stage applies to the outcome  $x$  of the first stage an *upwards probabilistic shift*. That is, if outcome  $x$  is realized in the first stage, then the second stage pays a final amount of money  $x + z$ , where  $z$  is distributed according to  $H_x(z)$  with  $H_x(0) = 0$ . Thus  $H_x$  generates a final return of at least  $x$  with probability one.

Denote the resulting distribution by  $F$ . Then for any nondecreasing  $u : Z \rightarrow \mathbb{R}$  we have

$$\int u(t) dF(t) = \int \left( \int u(x + z) dH_x(z) \right) dG(x) \geq \int u(t) dG(t).$$

Thus,  $F \triangleright_{\text{fosd}} G$ .

- We want now to introduce a comparison based on relative riskiness or dispersion.

### Definition 4.16

For any two distributions  $F$  and  $G$  with the same mean,  $F$  **second-order stochastically dominates**  $G$ , denoted  $F \triangleright_{\text{sosd}} G$ , if for every non-decreasing concave function  $u : Z \rightarrow \mathbb{R}$ , we have

$$\int u(z) dF(z) \geq \int u(z) dG(z)$$

### Definition 4.17

For any  $F, G \in \mathfrak{F}$ ,  $G$  is a mean-preserving spread of  $F$  if and only if

$$Y = X + \epsilon$$

for some random variables  $X \sim F$ ,  $Y \sim G$  and  $\epsilon$  such that  $E[\epsilon|X = x] = 0$  for all  $x$ .

- Imagine that for every realization of  $x$ , we add a noise  $\epsilon$  and give the decision maker  $y = x + \epsilon$ . In other words, we are spreading the probabilities without changing the mean.
- If the decision maker is risk averse, he would not like to have this scheme. He would rather have  $x$ .
- It is instructive to compare this to example 4.15. In that example, we were giving the decision maker an extra amount of money for every realization  $x$ .
- While this could increase the variance of money, the decision maker knew that he was getting more.
- In the current scheme, we are increasing the variance without increasing the expectation.

### Proposition 4.18

Assume that  $F, G \in \mathfrak{F}$  have the same mean. Then, the following are equivalent.

- ① For every non-decreasing concave function  $u : Z \rightarrow \mathbb{R}$ , we have  $\int u(z) dF(z) \geq \int u(z) dG(z)$
- ②  $G$  is a mean-preserving spread of  $F$ .
- ③ For every  $t \geq 0$ ,  $\int_w^t G(x) dx \geq \int_w^t F(x) dx$ .

### Proof.

Homework! You can use a similar technique as in the proof of 4.14, by using integration by parts twice. □

- Let  $F, G \in \mathfrak{F}$  be distributions with densities  $f$  and  $g$  respectively.
- Define the **odds-ratio**

$$l(x) = \frac{f(x)}{g(x)}.$$

- Suppose that you observe a draw  $x$  from a random variable that may have distribution  $F$  or  $G$ .
- You ask the question: *given the realization  $x$ , is it more likely that it came from  $F$  than from  $G$ ?*
- It turns out that  $l(x)$  is the ratio to which you need to multiply the prior odds to get the posterior odds!

### Definition 4.19

The pair  $(f, g)$  has the MLRP (monotone likelihood ratio property) if  $I(x)$  is non-decreasing.

- Intuitively, the higher the realized value of  $x$ , the more likely that it was drawn from the high distribution  $F$ .
- MLRP implies FOSD, but it is a stronger condition.
- You could have FOSD and still there might be some high signal values that likely come from  $G$ .

### Example 4.20

Suppose  $f(0) = f(2) = 0.5$  and  $g(1) = g(3) = 0.5$ . Then  $G \succ_{\text{fbsd}} F$  but the MLRP property fails.

- The MLRP is often used as an assumption in models of moral hazard and screening.
- Let  $F(x|a)$  be a family of distribution functions parameterized/indexed by the parameter  $a$ .
- The parameter  $a$  could be an action (e.g. effort), or a type (e.g. ability) of an agent, and  $x$  is the outcome (e.g. the amount produced/consumed).
- The MLRP tells us that if  $x_2 > x_1$  and  $a_2 > a_1$  then

$$\frac{f(x_2|a_2)}{f(x_2|a_1)} \geq \frac{f(x_1|a_2)}{f(x_1|a_1)}.$$

- In other words, if a principal observes a higher  $x$ , it will guess a higher likelihood that it came about due to a higher  $a$ .
- This is important in the design of incentives!